DRUGS · THYROID AND PARATHYROID

Recombinant TSH

Recombinant human TSH (thyrotropin alfa) stimulates residual thyroid tissue and cancer cells to take up iodine and release thyroglobulin without stopping levothyroxine. It spares patients the weeks of hypothyroidism that withdrawal causes, and is used to prepare for remnant ablation and for diagnostic scans and stimulated thyroglobulin testing.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

RECOMBINANT TSH

	SUMMARY
Agent	Thyrotropin alfa
How it works	Binds TSH receptors on thyroid cells, increasing iodine uptake and thyroglobulin release
ENT uses	Preparation for radioiodine remnant ablation in low and intermediate risk differentiated thyroid cancer; stimulated thyroglobulin and diagnostic whole-body scans in follow-up; patients who cannot tolerate hypothyroidism (heart failure, frailty, psychiatric illness) or cannot raise their own TSH (pituitary disease)
Dose	0.9 mg intramuscularly on 2 consecutive days; radioiodine on day 3; thyroglobulin measured on day 5
Cautions	Bulky residual or metastatic disease (TSH-stimulated tumour swelling can cause airway, spinal or brain compression; consider steroid cover)

2 CHAPTER TWO Using it in ENT

Practice.

2.1 Ablation and follow-up

Trials (HiLo, ESTIMABL) showed that ablation with low-dose radioiodine after recombinant TSH is as successful as after withdrawal in low-risk patients, with better quality of life. A stimulated thyroglobulin after recombinant TSH is used to assess response in some follow-up protocols, although highly sensitive thyroglobulin assays have reduced the need.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Nausea, headache, fatigue, flu-like symptoms; transient tumour swelling and pain at metastatic sites.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- The advantage over **withdrawal**.

- The dosing schedule and timing of radioiodine and thyroglobulin.

4.2 Common pitfalls

COMMON PITFALLS

- Giving recombinant TSH to a patient with tracheal or spinal metastases without steroid cover.

Written for this app from BNF and BNF for Children; NICE and specialist society guidance; Scott-Brown; Cummings. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.